Ali Saeed Chaudhary

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EDUCATION

Iowa State University Ames, Iowa
B.S., Mechanical Engineering, Cyber-Physical Systems Minor. ***GPA: 4.0***. *May 2028*
Korea University Seoul, South Korea
Study Abroad Program. *June 2026*
Relevant Coursework: Mechatronics, Reinforcement Learning with Robotics Applications, Nuclear Power

EXPERIENCE

Research Assistant Aug 2026 – Present
Iowa State University, Coordinated Systems Lab *Ames, Iowa*
- Developing RNN-based models to predict deviations from planned aircraft trajectories and estimate forecast uncertainty, in collaboration with RTX Collins Aerospace.
- Completed Python preprocessing pipelines for the NASA Sherlock database, aligning flight-plan and recorded-trajectory data to prepare trajectory-deviation targets for model training.

Radiology IT Support Coordinator Nov 2024 – Feb 2026
Concord Medical Management, LLC *Flower Mound, Texas (remote)*
- Acted as primary point of contact for radiologists and healthcare facilities, diagnosing and resolving time-sensitive technical issues affecting medical imaging workflows.
- Owned troubleshooting and issue escalation, coordinating directly with facilities to restore system functionality.

Lab Engineering Intern May 2024 – Aug 2024
LUMS FMOD Research Lab *Lahore, Pakistan*
- Led design and construction of a compact substrate-rubbing machine with a custom-milled aluminum frame, Arduino Mega/RAMPS 1.4 control system driving three NEMA 17 stepper motors.
- Designed the mechanical assembly in Onshape, developed a custom control-panel PCB, and 3D-printed control surfaces and button caps for automated operation and manual adjustment.
- Implemented an automated rubbing routine with roller retraction and limit-switch-based stage homing; deployed the completed machine to the lab.
- Built a PID-controlled temperature-regulated microscope stage as a secondary project during the same internship.

ACTIVITIES & LEADERSHIP

Cardinal Space Mining, NASA Lunabotics Aug 2026 – Present
Collections Sub-Team Member
- Leading dust-cover design for the collections sub-system of a lunar regolith mining rover
- Designing to improve robustness over the prior year's design while maintaining functionality and manufacturability.

Cyclone Rocketry, sponsored by Boeing Oct 2024 – Dec 2025
Mechanical Sub-Team Member
- Proposed and led design of a rocket airbrake mechanism which was installed and launched in the club's rocket
- Modeled components in SolidWorks and ran FEA simulations in Ansys.

PROJECTS

RL-Trained Robotic Hand May 2026 – July 2026
- Trained a reinforcement learning policy in Isaac Sim to control a robotic hand using reference data from the HO-Cap (NeurIPS 2025) dataset.
- Engineered a custom reward function prioritizing fingertip and palm contact, stable grasp formation, and reduced unnecessary motion.
- The trained model successfully grabbed and manipulated objects in Isaac Sim.

Private Home Server June 2026 – Present
- Building a private media server for storing, hosting, and distributing services and files.
- Configured remote access through an Oracle VPS using Pangolin and Traefik as a reverse proxy, routing multiple services behind a single entry point.

SKILLS

CAD & Simulation: SolidWorks, Onshape, Fusion 360, Isaac Sim, Ansys FEA, CAM
Manufacturing: CNC Milling, Welding (TIG, MIG), Metal Working, 3D-Printing
Programming & ML: Python, RNNs, PPO, SAC, VLM / VLA Models, Robot OS, Arduino, STM32, PID Control